

CHALLENGER

CONCERNED WITH drumming up public support for the Shuttle program, NASA announced in 1985 that a teacher would be selected for a space flight. More than 11,000 applications were received from teachers across America. Christa McAuliffe of Concord, New Hampshire, won the competition amid a blaze of publicity that was a triumph for the space agency's public relations.

McAuliffe joined the crew for a flight on board *Challenger* scheduled for January 1986, along with mission commander Dick Scobee,

CHALLENGER CREW

The crew of the Shuttle *Challenger* pose cheerfully for the camera before their ill-fated flight. High school teacher Christa McAuliffe is second from the left in the back row.

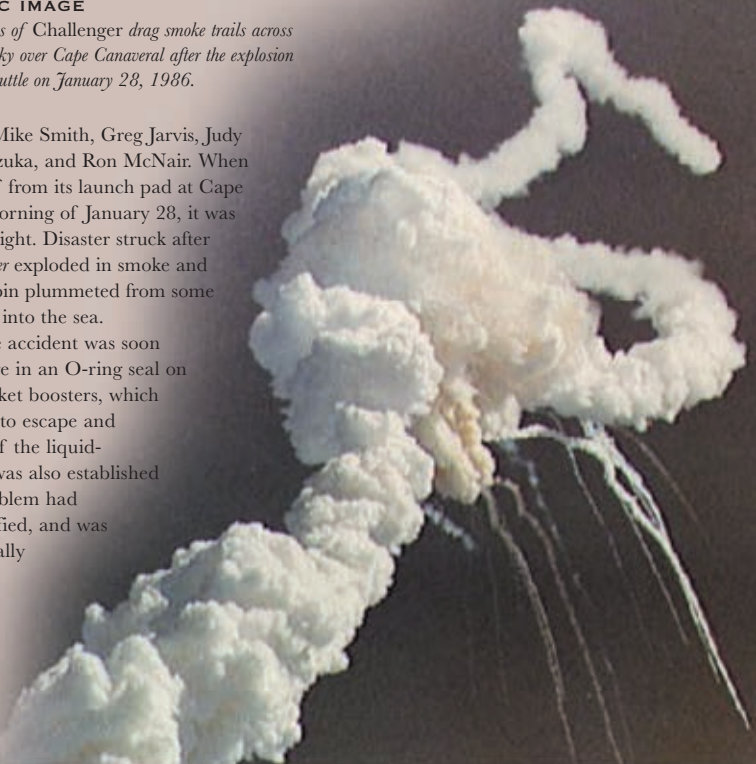


TRAGIC IMAGE

Fragments of *Challenger* drag smoke trails across the blue sky over Cape Canaveral after the explosion of the Shuttle on January 28, 1986.

Shuttle pilot Mike Smith, Greg Jarvis, Judy Resnik, Ellison Onizuka, and Ron McNair. When *Challenger* lifted off from its launch pad at Cape Canaveral on the morning of January 28, it was beginning its 10th flight. Disaster struck after 73 seconds. *Challenger* exploded in smoke and flames; the crew cabin plummeted from some 9 miles (14km) high into the sea.

The cause of the accident was soon identified as a failure in an O-ring seal on one of the solid rocket boosters, which had allowed flames to escape and penetrate the skin of the liquid-propellant tank. It was also established that the O-ring problem had already been identified, and was known to be especially dangerous in cold weather. The temperature on the morning of January 28 was below freezing.



at Edwards Air Force Base. The first Shuttle flight into space had originally been projected for 1979, but problems, especially with the main rocket engines, produced delays and helped push the program \$1 billion over budget. Nonetheless, on April 12, 1981, exactly 20 years after Gagarin's epoch-making space flight, the Shuttle *Columbia* lifted off from Cape Canaveral with John Young and Robert Crippen at the controls. Despite a nerve-racking moment when the astronauts discovered that some of the heat-protective tiles that would prevent the vehicle burning up on reentry had shaken loose, *Columbia* returned to the Earth's atmosphere without mishap and

smoothly glided to land at Edwards AFB. Further test flights confirmed the success of the Shuttle, which went operational in November 1982.

Technologically, the Shuttle was one of the wonders of the world. But as a program that had promised to make space flight routine and, in the words of President Richard Nixon, to "take the astronomical costs out of astronautics," it was soon in serious trouble. The Shuttle, it turned out, was neither cheap nor easy to operate. NASA had worked out costings on the basis of 24 Shuttle flights a year. Yet even with engineers and technicians working overtime, it proved difficult to achieve

a launch every two months. The Shuttle was supposed to cover its costs by luring paying customers who needed satellites launched or other space services. But even charging only a fraction of the real cost of a launch, the Shuttle was undercut by the European Space Agency's Ariane rockets. Even the USAF deserted the Shuttle as a launcher for reconnaissance satellites, after *Challenger*, reverting to disposable launch rockets.

NASA continued to promote the Shuttle as above all a safe means of travel – a spaceliner usable by anyone. For the first time, people who were not pilots voyaged into space, first scientists or engineers who conducted experiments on board, and then selected passengers whose presence might boost the program. One of these, teacher Christa McAuliffe, was on board when *Challenger* exploded on January 28, 1986. Further Shuttle missions were postponed as an investigation began into the causes of the accident. It was NASA's darkest hour.

SHUTTLE PIGGYBACK

A Boeing 747 was adapted to transport the spacecraft from its landing strip at Edwards Air Force Base, California, to its launch point at Cape Canaveral, Florida. Initial flight tests of the Shuttle's gliding ability were also carried out by air launch from the back of the 747.

